Attention for Multi-Excitation Fluorescence Hyperspectral Pixel Classification: A Leakage-Free Evaluation

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***Abstract*—Multi-excitation (ME) fluorescence hyperspectral imaging records, for every pixel, an emission spectrum under several excitation wavelengths, i.e. a sparse and irregularly sampled excitation–emission matrix (EEM). Pixel classifiers for such data are usually evaluated with random pixel splits, which place neighbouring pixels of the same specimen in training and test sets and overstate accuracy. We study supervised pixel classification of ME images under a strict whole-specimen protocol: every test fold contains complete, spatially disjoint specimens of every class, hyper-parameters are tuned on held-out training specimens, and all models receive the same preprocessing, training procedure and tuning budget. We propose a family of attention classifiers for irregular EEM data with continuous wavelength encodings (measurement tokens, excitation tokens, a convolutional-stem hybrid, and a spatial extension that attends over a pixel neighbourhood). On three datasets (Collagen Pepsin, Lichens 1, Lichens 2; 3–5 classes, 158–210 measurements per pixel) and two tracks (spectral-only and spectral–spatial), we compare it with 13 tuned baselines (8 spectral, 5 spatial), including SpectralFormer, SSFTT, and tree, kernel and CNN classifiers. Tokenising whole emission spectra per excitation (excitation tokens) is the best model on Collagen Pepsin ($91.1 \pm 7.6\%$ balanced accuracy) and within 0.1 points of the best baseline on Lichens 2, and it is better than SpectralFormer on Pepsin; but it is not reliably better than the strongest tuned baseline on any dataset (logistic regression, RBF SVM and SpectralFormer, respectively; paired differences -0.6 to $+0.8$ points, confidence intervals including zero on Pepsin and Lichens 2). Tokenising single measurements, a prototype and a convolutional-stem hybrid are 3 to 11 points below the best baselines on Pepsin and Lichens 2. Attention over measurement tokens selects the same bands as an unsupervised band-selection method far more often than chance (13 of 30 bands on Pepsin, hypergeometric $p = 0.0005$). On a development set, random-pixel splits give 98% balanced accuracy for several models, compared with 87–88% under whole-specimen evaluation, an effect larger than the differences between the best models. We release code and the evaluation protocol.**

***Index Terms*—Hyperspectral imaging, fluorescence, excitation–emission matrix, attention, transformer, evaluation protocol, data leakage.**

I. Introduction

MULTI-EXCITATION (ME) fluorescence hyperspectral imaging (HSI) extends single-excitation fluorescence imaging by illuminating a specimen with several excitation wavelengths in turn and recording a full emission spectrum for each. The result is a four-dimensional array (two spatial axes, excitation, emission). Per pixel it is an excitation–emission matrix (EEM) of the kind long used in fluorescence spectroscopy, where the joint dependence on excitation and emission carries information that a single excitation does not [1], [2]. Endogenous fluorophores in tissue, such as collagen, NADH and elastin, differ in both their excitation and emission profiles [3], [4], and ME-HSI has been used to classify tissue states and to delineate lesions [5]–[7]. Medical HSI more broadly is reviewed in [8].

In the setting of this paper, a tunable light source provides 6–8 excitation wavelengths between 310 and 430 nm and a liquid-crystal tunable filter (LCTF) camera records emission between 420 and 750 nm in 10 nm steps. The resulting pixel is not a regular $E \times M$ matrix. Each excitation has its own emission window: the lower bound rises with the excitation wavelength (from 420 nm to 470 nm at 430 nm excitation in the lichen data), the upper bound varies between 630 and 750 nm, and for the shorter excitations of the collagen and lichen data a contiguous block of 70–90 nm of bands is absent [confirm reason for the removed block, e.g. excitation or second-order light]. In the lichen data the 415 nm excitation is sampled on a grid offset by 5 nm from the others. A classifier that treats the 150–210 measurements as an unordered, unlabelled feature vector discards the wavelength structure; one that assumes a regular grid must either interpolate or pad. Two natural questions follow. Does a model that knows the excitation and emission wavelength of each measurement classify better than one that does not? And does attention, which can relate measurements across excitations without a fixed layout, offer an advantage over convolutional, kernel and tree-based classifiers once all are tuned equally?

The second question cannot be answered reliably with the evaluation common in the HSI classification literature. Most benchmark protocols draw training and test pixels at random from the same image. Because neighbouring pixels are strongly correlated, such splits measure interpolation within a scene, not generalisation to new scenes or specimens, and the resulting accuracies are inflated [9], [10]. The same problem is known in other fields under the names spatial autocorrelation and data leakage [11], [12]. For ME-HSI of biological specimens the relevant unit of generalisation is the specimen (a tissue sample, a lichen thallus), and the relevant error is the one made on a specimen that was never seen during training. In a preliminary analysis of our own earlier prototype we found two further sources of optimistic or unstable numbers: emission bands were aligned by index instead of by wavelength, and fold assignment did not guarantee that each test fold contained every class (Sec. V-A). These issues are easy to introduce and hard to see in a single reported accuracy, which motivates a

Manuscript in preparation. [funding / acknowledgements]

fixed, pre-specified protocol.

Contributions. (i) A leakage-free benchmark protocol and harness for ME-HSI: whole-specimen, class-covering nested cross-validation with equal tuning budgets and shared preprocessing and training for all models, and statistics with corrections for multiple comparisons. (ii) An attention family for irregular EEM data with continuous wavelength encodings: measurement tokens, excitation tokens, a convolutional-stem hybrid, and a spatial extension. Axial attention, relative-bias and gated variants were implemented but not evaluated in the final sweep. (iii) An evaluation on three datasets and two tracks against 13 tuned baselines. Excitation-level tokenisation is competitive with the best tuned baselines and better than the other neural models, including SpectralFormer, on one dataset (Collagen Pepsin); tokenising single measurements is not. We report a negative result: no attention variant has a reliable advantage over the strongest tuned linear or kernel baseline. (iv) Evidence that the protocol matters more than the model: random-pixel splits inflate balanced accuracy by 10–12 points on a development set, more than the difference between the best models. (v) A descriptive comparison showing that measurement-token attention importance agrees with the wavelengths selected by an unsupervised method [13] far above chance on two datasets.

II. Related Work

A. Deep learning for HSI classification

Early deep models for HSI classification used 1D convolutions over the spectrum, 2D convolutions over spatial patches, and 3D convolutions over both [14], [15]; a comparative review is given in [16]. HybridSN stacks 3D and 2D convolutions to combine spectral and spatial features at moderate cost [17]; we implemented it but did not include it in the final comparison, where a 2D-CNN and SSFTT represent convolutional spectral–spatial models. Transformers [18], [19] entered the field through SpectralFormer, which embeds groups of neighbouring bands as tokens and adds cross-layer feature fusion [20], and SSFTT, which builds tokens from 3D–2D convolutional features with a Gaussian-weighted tokenizer [21]. Foundation models such as SpectralGPT are pre-trained on large collections of spectral remote sensing images [22]. These models assume that the spectral axis is a regular, ordered sequence of broad-band or narrow-band channels, and positional information is learned or implied by index. ME fluorescence data do not fit this assumption: the measurements are indexed by two wavelengths, the grid is irregular, and neighbouring indices need not be neighbouring wavelengths. We therefore include SpectralFormer and SSFTT as baselines adapted to our feature layout, and design the attention family around wavelength-valued, not index-valued, positions.

B. Fluorescence EEM analysis

EEMs are classically analysed with multi-way decompositions such as PARAFAC, which factor a stack of EEMs into excitation and emission loadings under a trilinear model [1], [2]. These methods are unsupervised, assume a regular grid after correction of scatter and missing regions, and operate on samples (or on pixels as samples) in aggregate. Our setting is supervised and per-pixel, and the model must tolerate missing and irregular cells.

C. Evaluation leakage in HSI classification

Liang *et al.* showed that random sampling of training pixels in scenes with spatially extended classes produces over-optimistic accuracy for spectral–spatial methods, because training and test patches overlap, and argued for spatially disjoint sampling [9]. Nalepa *et al.* reached a similar conclusion for HSI segmentation and proposed leakage-free validation [10]. General treatments of cross-validation for structured data are found in [11], and a survey of leakage in ML-based science in [12]. Our protocol follows this line and enforces three additional constraints: folds hold out complete specimens, every test fold contains every class, and model selection uses held-out specimens inside the training set, in the spirit of nested cross-validation [23].

D. Generalisation across specimens and domains

Specimen-to-specimen variation (preparation, thickness, exposure, ageing) is a form of domain shift. Domain generalisation methods address it with invariance objectives, data augmentation and regularisation [24], [25], and a careful comparison found that tuned empirical risk minimisation is a strong baseline when model selection and budget are controlled [26]. We adopt the second lesson as a design constraint: every model, ours and the baselines, gets the same tuning budget and model-selection rule.

III. Data

A. Acquisition

Specimens were imaged with a tunable-source ME fluorescence system. For each excitation wavelength, an emission cube was recorded with an LCTF camera at 10 nm steps from 420 nm up to 720 nm (collagen, Lichens 1) or 750 nm (Lichens 2). Exposure time and lamp power were recorded per excitation and used to put measurements on a common radiometric scale before analysis [confirm radiometric correction applied to each dataset]. Excitation wavelengths are 310, 325, 340, 365, 385 and 400 nm for the collagen data and additionally 415 and 430 nm for the lichen data. Each excitation therefore keeps its own set of emission bands (windows of 24–31 bands per excitation in the collagen data and 22–30 in the lichen data), and the cube is ragged. In both lichen datasets the 415 nm excitation is sampled at 455, 465, … nm, i.e. in 10 nm steps on a grid offset by 5 nm from the other excitations, which gives union emission grids of $M = 55$ (Lichens 1) and $M = 63$ (Lichens 2) distinct wavelengths for 8 excitations. Spatial size is 256×348 pixels for the collagen data and Lichens 2, and 1040×925 for Lichens 1.

B. Datasets and specimens

Class labels are given by a region mask drawn on the image. A *specimen* is an 8-connected component of one class in the mask; components with fewer than 100 pixels are discarded (none for Collagen Pepsin and Lichens 1, one component of 18 pixels for Lichens 2, and one of 11 pixels for the development set). For the collagen and Lichens 1 data the specimens are physically separate samples; for Lichens 2 they are spatially disjoint regions that may belong to the same thallus (Sec. VIII). Table I summarises the data.

C. Irregular emission grids

Models must not assume a regular $E \times M$ grid. All loaders align measurements by their wavelength in nanometres and refuse to fall back to index alignment when a cube lacks a wavelength vector. Features are stored as a flat list of valid measurements sorted by (excitation, emission), with the arrays λ_i^{ex}, λ_i^{em} and the grid position (e_i, m_i) of each feature. Models that need a grid scatter the features onto $E \times M$ and carry a validity mask.

IV. Method

A. Problem formulation

Let a pixel be a set of F valid measurements $\{(\lambda_i^{\mathrm{ex}}, \lambda_i^{\mathrm{em}}, x_i)\}_{i=1}^{F}$ with $x_i \geq 0$ the recorded intensity. In the spectral track the classifier $f : \{(\lambda_i^{\mathrm{ex}}, \lambda_i^{\mathrm{em}}, x_i)\} \to \Delta^{C-1}$ maps one pixel to class probabilities. In the spatial track it receives a $P \times P$ patch of such sets centred on the pixel, with pixels outside the centre pixel's specimen set to zero, so that no information crosses specimen boundaries. Both tracks are trained with labelled pixels from training specimens and evaluated on pixels of unseen specimens.

B. Measurement-set representation

Intensities are preprocessed with a shared, trainable-free transformer (Sec. V-D); the model sees standardised features $\tilde{x}_i$. Each measurement is a token, so the number of tokens is the number of valid cells and no padding or interpolation is needed. Wavelengths enter as continuous quantities, not indices.

C. Attention family

Fig. 1 shows the measurement-token model, one member of the family, and Fig. 2 shows the excitation-token model and the hybrid variant. The components below are options; their configuration is chosen by inner validation (Sec. V) within the frozen search spaces. Four spectral members were evaluated: the corrected prototype, measurement tokens, excitation tokens and the conv-stem hybrid.

1) Measurement tokens with continuous wavelength encoding: Token i is $\mathbf{t}_i = \phi(\tilde{x}_i) + \psi(\lambda_i^{\mathrm{ex}}, \lambda_i^{\mathrm{em}}, s_i)$ with Stokes shift $s_i = \lambda_i^{\mathrm{em}} - \lambda_i^{\mathrm{ex}}$. The value embedding ϕ is a learned scale-and-shift vector or a small MLP. The wavelength encoding ψ rescales the three quantities to $[0, 1]$ and applies Fourier features [27]: for $u \in [0, 1]$, $\gamma(u) = [u, \sin(2\pi k u), \cos(2\pi k u)]_{k=1}^{K}$, followed by a linear map to the model width. A learned per-feature embedding is available as an alternative. A classification token is prepended, or tokens are pooled by a learned-query attention (searched: Fourier or Fourier-plus-learned encoding, with or without relative bias, CLS or attention pooling). With 150–210 tokens this is the most expensive member. The encoder is a pre-norm transformer [18] with stochastic depth and optional random dropping of measurement tokens during training.

2) Relative wavelength bias: Following relative position representations [28] and learned relative biases [29], attention logits receive an additive per-head bias $b_h = \mathrm{MLP}_h(|\Delta\lambda^{\mathrm{ex}}|, |\Delta\lambda^{\mathrm{em}}|, |\Delta s|)$, with distances normalised by the corresponding wavelength range. The last layer is zero-initialised, so training starts from plain attention. The bias depends on physical distance, so it transfers across excitation sets and irregular grids.

3) Axial attention on the EEM (implemented, not evaluated): Tokens are placed on the $E \times M$ grid and blocks alternate attention along emission (within one excitation) and along excitation (at one emission wavelength), with key-padding masks for missing cells. This restricts attention to the two axes along which EEMs are physically structured and reduces cost from $O(F^2)$ to $O(EM^2 + ME^2)$. It was implemented and gave a competitive result on the development set (85.3%) but is not part of the final sweep, for cost and redundancy with the excitation-token model.

4) Excitation tokens: The excitation-token model (`ex_tokens_v2`, Fig. 2) forms one token per excitation from that excitation's emission spectrum. The F features are scattered onto the $E \times M$ union grid, missing cells are set to zero, and a binary validity mask is a second channel. The emission axis is zero-padded to a multiple of $2S$ and split into S contiguous chunks ($S = 1$ by default, i.e. one token per excitation, or $S = 2$ sub-tokens). Each chunk, with its mask, is embedded by a single linear layer (default) or by a small 1-D convolution (two k=5 layers, 16 channels, input channels value, mask and normalised emission wavelength) followed by a linear layer. The token is the sum of this embedding, a Fourier encoding of the pair (excitation wavelength, chunk-centre emission wavelength) with K=8 frequencies, and a learned chunk embedding. Tokens whose chunk contains no valid cell are masked as keys, and during training further tokens are dropped at random (probability 0.1). A classification token is prepended and a pre-norm transformer encoder (default width 64, depth 3, 4 heads, MLP ratio 2) acts on the E (or ES) tokens; the classification token feeds a linear classifier (mean pooling over valid tokens is available). The searched options are width $\{32, 64\}$, depth $\{2, 3\}$, sub-tokens $\{1, 2\}$ and dropout $\{0.1, 0.3\}$; the linear embedding and CLS pooling are fixed. Cross-excitation interactions are explicit and the sequence is 8–16 tokens long, so the model is much cheaper than measurement tokens.

5) Hybrid convolutional-stem attention: `hybrid_conv_attn` replaces the chunk embedding by a learned local feature extractor. A stem of two 1-D convolutions (k=5, 24 channels, GELU; input channels value, mask and normalised emission wavelength) is shared

TABLE I

DATASETS. S_c: SPECIMENS (DISJOINT REGIONS) PER CLASS; E: EXCITATIONS; M: DISTINCT EMISSION WAVELENGTHS IN THE UNION GRID; F: VALID (EXCITATION, EMISSION) MEASUREMENTS PER PIXEL. COLLAGEN ACETIC ACID IS THE DEVELOPMENT SET AND IS NOT USED FOR ANY REPORTED TEST NUMBER.

Dataset	Classes	S_c	Pixels	Excitations (nm)	E	M	F	Emission (nm)
Collagen Pepsin	3	3/3/3	39 970	310–400	6	31	158	420–720
Lichens 1	4	4/4/4/4	191 046	310–430	8	55	192	420–720
Lichens 2	5	4/10/7/5/2	22 562	310–430	8	63	210	420–750
Collagen Acetic (dev.)	3	3/3/3	46 670	310–400	6	31	149	420–720

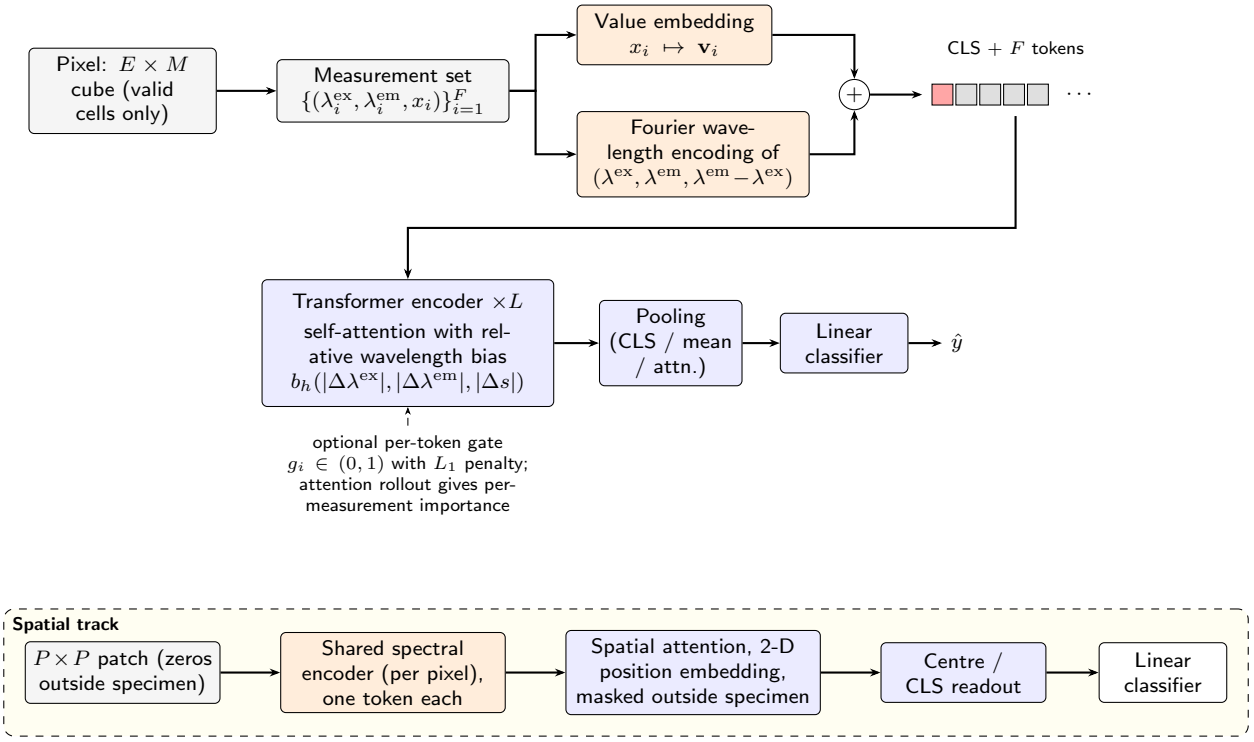


Fig. 1. Measurement-token attention model (top) and its spatial extension (bottom). A pixel is a set of (excitation, emission, intensity) measurements. Each token is the sum of a value embedding and a Fourier encoding of $(\lambda^{\mathrm{ex}}, \lambda^{\mathrm{em}}, \lambda^{\mathrm{em}} - \lambda^{\mathrm{ex}})$. A transformer encoder with a learned relative wavelength bias and optional token gates produces a pooled representation for a linear classifier. In the spatial track a shared spectral encoder maps every patch pixel to a token and a second attention stage reads out the centre pixel.

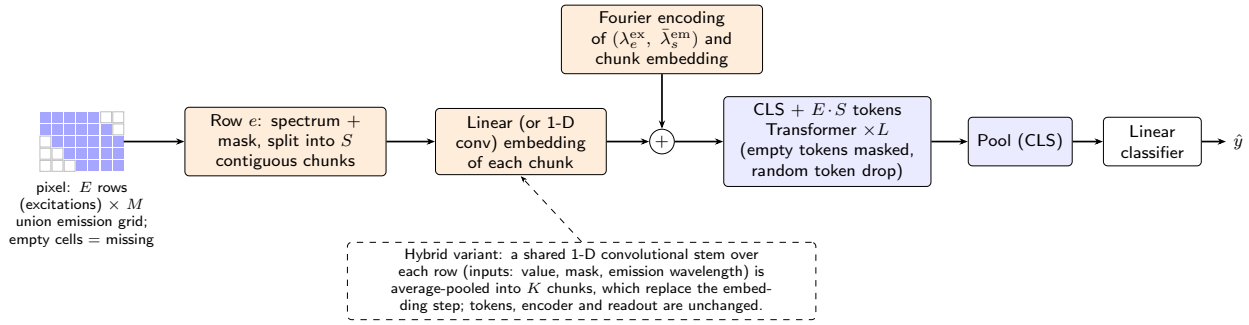


Fig. 2. Excitation-token model. Each excitation's emission spectrum on the union grid (with a validity mask) is split into S chunks, embedded, and combined with a Fourier encoding of the excitation and chunk-centre emission wavelengths; a transformer over the $E \cdot S$ tokens and a CLS readout give the prediction. The hybrid variant (dashed) replaces the embedding by a shared 1-D convolutional stem and pooling into K chunks.

across excitations and applied along the emission axis of each excitation row. Its output is average-pooled into K contiguous chunks of the union grid, projected to the model width, and given the same Fourier encoding of (excitation, chunk-centre emission) as above, giving $E \cdot K$ (excitation, chunk) tokens. A CLS token, a transformer encoder (default width 64, depth

2, 4 heads) and a linear classifier follow, with the same token masking and token dropout as the excitation-token model. The searched options are width $\{32, 64\}$, depth $\{1, 2, 3\}$, chunks $K \in \{4, 6, 8\}$ and dropout $\{0.1, 0.3\}$.

6) Gated tokens (implemented, not in the frozen search space): A learned gate $g_i = \sigma(\theta_i)$ per measurement scales the token and is added as $\log g_i$ to the attention logits of that key. An L_1 penalty on the mean gate encourages sparsity, and the gate values give a ranking of measurements for the analysis in Sec. VI-D. Attention rollout [30] provides a second, gate-free importance measure; we treat both as descriptive, not causal [31].

7) Reference prototype: The prototype (`smartattention_v0`) is a SpectralFormer-style model in which tokens are emission positions carrying the intensities of all excitations and their two spectral neighbours, with cross-layer feature fusion. It is evaluated on the aligned wavelength grid (after the grid-alignment correction of Sec. V-A) and is included as a model, not as a baseline, to show the effect of the more general tokenisations.

D. Spatial extension

For the spatial track a shared spectral encoder (an excitation-token encoder or an MLP, chosen by the search) maps each valid pixel in the $P \times P$ patch to one token. A 2-D learned position embedding is added, pixels outside the centre pixel's specimen are masked as keys, and a second transformer attends over the pixel tokens. The prediction is read from a classification token or from the centre token, optionally concatenated with the centre pixel's own spectral embedding. Patch size is tuned in $\{5, 7\}$.

E. Training objective

The final objective is plain label-smoothed cross-entropy (smoothing 0.05) with the shared training recipe of Sec. V-D; there is no invariance loss, no gate penalty in the evaluated models, and no contrastive or adversarial term. This follows from the development experiments below, in which none of these terms helped.

F. Development experiments

All results in this subsection are on the development set (Collagen Acetic Acid; 3 specimens per class, 6 (repeat, fold) units), whose standard deviations are 4–10 points, so differences below about 3 points are not resolved. They were used to fix the preprocessing space and the shared training options, not to select among final models. *Per-excitation normalisation hurts.* Normalising each excitation block separately lowered balanced accuracy by 4–8 points for every model family (MLP 73 vs 81%, logistic regression 77 vs 81%, gradient boosting 73 vs 81%), because the cross-excitation intensity ratios are class-informative. The final search space therefore contains only global normalisation. *Invariance-oriented objectives did not help.* Per-excitation intensity-scale augmentation (3–10 points worse), log-centring of intensities per pixel or per excitation (3–8 points worse), supervised contrastive loss with cross-specimen positives ($+1.5$ to -0.6) and conditional specimen-adversarial training (-0.9) gave no gain beyond noise or hurt; same-class mixup was not completed. We therefore do not use them. *Excitation dropout.* Zeroing whole excitation blocks gave $+2.3$ points for the MLP (5 of 6 units) and $+0.4$ for the hybrid attention model, i.e. the gain was not specific to attention; it is a shared option for all neural models in the final sweep. *Attention versus MLP.* On the development set the best attention variants tied a well-trained MLP (86.5 for the hybrid model with scale-and-noise augmentation vs 86.7% for the MLP, std 4–10 over 6 units), so the dev data give no evidence of an attention advantage.

V. Evaluation Protocol

The protocol was fixed before any outer-test result on the final datasets was produced; search spaces were frozen first. Collagen Acetic Acid, the development set, was used for architecture exploration with its own full cross-validation; the final datasets contributed only inner-validation scores during development.

A. Motivation: what the prototype got wrong

Our earlier prototype aligned emission bands by index, which placed 31 real wavelengths onto 30 positions on the development set, and drew test specimens at random, so that on seed 42 two of three test folds were missing a class (test-fold classes $[1, 1, 2]$ and $[0, 0, 2]$). After alignment and class-covering folds, accuracy of the same model class changed (whole-specimen balanced accuracy 79.8–81.5% before and 81.7–82.7% after for the three edge modes; logistic regression 63.7% before and 73.8% after) and the fold-to-fold range narrowed. Random-pixel splits on the same data (3 folds, default configurations, balanced random pixel split with pixels of every specimen in train and test) gave 97.9% (logistic regression), 98.4% (MLP) and 98.4% (prototype), compared with 87.8, 86.8 and 88.3% for whole-specimen folds, a gap of 10–12 points. On the development set, five different fold assignments gave balanced accuracy of $81.1 \pm 0.8\%$ (attention, fixed configuration) and $76.2 \pm 2.1\%$ (logistic regression), which is the scale of variation any comparison needs to exceed.

B. Outer folds

Let S_c be the number of specimens of class c and $k = \min_c S_c$. For repeat $r = 0, \ldots, R-1$ the specimens of each class are shuffled with seed r and dealt round-robin into k folds, so every fold contains at least one specimen of every class; this is asserted in code. Repeats were limited by compute (see Sec. VIII). The sweep started with $R = 2$ for Pepsin ($k = 3$, 6 units), $R = 2$ for Lichens 2 ($k = 2$, 4 units) and $R = 1$ for Lichens 1 ($k = 4$, 4 units). After seeing the first two repeats of the spectral track we extended the spectral models to $R = 5$ on Pepsin (15 units) and Lichens 2 (10 units) to gain statistical power; the extension was applied equally to all 12 spectral models, and Lichens 1 was not extended (accuracies near ceiling, highest cost). Spatial-track models

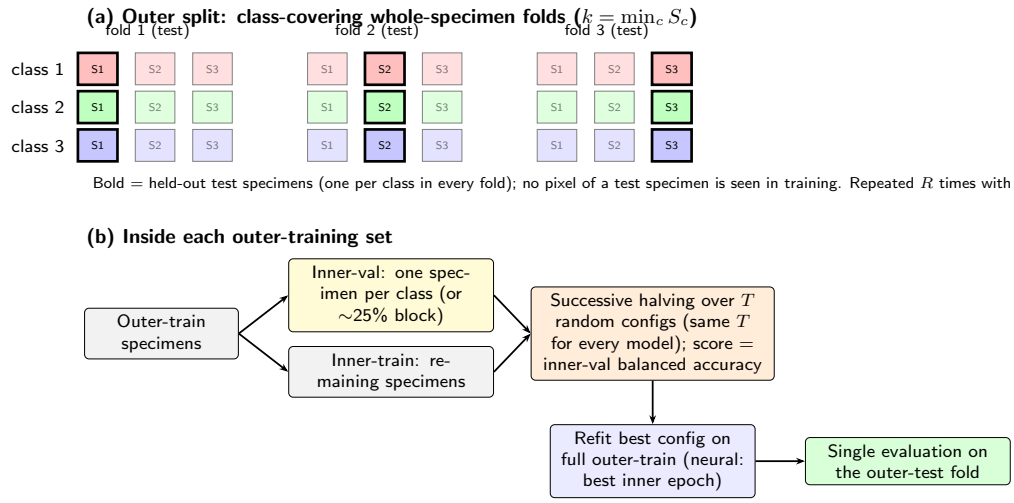


Fig. 3. Evaluation protocol. (a) Outer folds hold out whole specimens and always contain at least one specimen of every class. (b) Inside each outer-training set, one specimen per class is held out for model selection; successive halving selects a configuration with the same budget T for every model, which is refit on all training specimens and evaluated once on the test fold.

keep the initial 6, 4 and 4 units. All comparisons are between models run on identical outer folds. Test pixels of one fold belong to specimens that do not appear in the training set.

C. Inner validation and tuning

Within each outer-training set, one specimen per class is held out as inner-validation (chosen at random with a seed); if a class has a single training specimen, a spatially contiguous 25% block along the longer axis of that specimen is held out instead. Every model is tuned by random search over its own search space with the *same* budget $T = 3$ configurations (the default configuration is candidate 0 and two are drawn at random). For neural models, selection uses successive halving [32], [33]: all T candidates are trained for $\lceil E_{\max}/5 \rceil$ epochs, the better half is trained for $\lceil E_{\max}/2 \rceil$ epochs and the better half of those for $E_{\max} = 25$ epochs (5, 13 and 25 epochs; early stopping with patience 6 on inner-validation balanced accuracy). The planned larger budget ($T = 8$, $E_{\max} = 40$, more repeats) was reduced because of compute constraints (Sec. VIII). Classical models are fit for each of the T configurations. The score is inner-validation balanced accuracy. The best configuration is refit on all training specimens (neural models: for the best inner epoch, at least three) and evaluated once on the test fold. Random search is used for its efficiency over grids [34]; nesting avoids selection bias in the reported estimate [23]. Equal budgets follow [26].

D. Shared preprocessing and training

Preprocessing is part of every model's search space, with the same options for all: per-pixel normalisation $\in$ {none, L_2, area} applied across all features (global scope; per-excitation scope was dropped after the development experiments), an optional $\log(1+x/s)$ transform (s the median positive training value), and per-feature standardisation. Statistics are fitted on training pixels only. All neural models, baselines and ours, share one training loop: AdamW [35] with linear warm-up (3 epochs) and cosine decay, weight decay 10^{-4}, gradient-norm clipping, label smoothing 0.05, and specimen-balanced sampling with replacement (500 pixels per specimen and epoch), which gives each training specimen the same expected weight regardless of its pixel count. Shared options are the same for all neural models: augmentation $\in$ {none, scaling plus noise} (log-uniform scale in $[0.7, 1.4]$, an exact no-op whenever normalisation is on, plus Gaussian noise of 1% of the feature standard deviation), excitation dropout $\in \{0, 0.25\}$ (whole excitation blocks zeroed after preprocessing), and learning rate $\in \{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$. Early stopping monitors inner-validation balanced accuracy. Classical models use the same preprocessing and specimen-balanced sample weights (or specimen-balanced subsampling where the method has no weights).

E. Baselines

Spectral track (8): logistic regression, RBF SVM, random forest, histogram gradient boosting, k-NN, MLP, 1D-CNN over the (excitation, emission)-sorted feature vector, and pixel-wise SpectralFormer [20]. *Spatial track (5):* logistic regression and random forest on the concatenation of the centre spectrum and the patch-mean spectrum, 2D-CNN [14], [15], patch-wise SpectralFormer [20], and SSFTT [21]. Deep spatial baselines are scaled down to the 150–210 features and 5–7 pixel patches of our data. Patch size is $P \in \{5, 7\}$; to bound cost, training and inner-validation pixels are capped at 2000 per specimen (a fixed random subset), while test folds are evaluated on all pixels. Linear SVM, extra trees, a band-token vision transformer, a 3D-CNN and HybridSN [17] were implemented but dropped from the final sweep for cost and redundancy with the retained models. Pre-trained foundation models such as SpectralGPT [22] were not run because their input assumptions (fixed band layout, remote-sensing pre-training) do not match ME fluorescence data.

F. Metrics

The primary metric is pixel-level balanced accuracy (mean per-class recall) on the pooled test pixels of a fold. We also report macro-F1, pixel accuracy, specimen-averaged accuracy (mean over test specimens of their pixel accuracy), specimen majority-vote accuracy, per-class recall and the confusion matrix. Labels are fixed to $0, \ldots, C-1$ so that metrics are comparable across folds.

G. Statistics

The unit of analysis is a (repeat, fold) pair. For each dataset and each baseline, we compare our best model per track with the baseline on paired differences of balanced accuracy: the mean difference, a paired Wilcoxon signed-rank test, the Nadeau–Bengio corrected resampled t-test [36] with statistic $t = \bar{d}/\sqrt{(1/n + n_{\mathrm{te}}/n_{\mathrm{tr}})s_d^2}$ to account for overlap between training sets, and a bootstrap 95% confidence interval of the mean paired difference (10 000 resamples over units). p-values are adjusted across baselines with the Holm procedure [37]. Datasets are reported separately and are not pooled; we summarise them by win/tie/loss counts. Rankings across models on one dataset are visualised with critical-difference diagrams following [38], with the caveat that units within one dataset are not independent.

VI. Results

All results are from the `night` sweep: $T = 3$ configurations per model, $E_{\max} = 25$ epochs, $R = 5$ repeats on Pepsin and Lichens 2 and $R = 1$ on Lichens 1 for the spectral models (15, 10 and 4 units), and 6, 4 and 4 units for the spatial models, on one Apple M4 Pro GPU (MPS) shared with other jobs. All (repeat, fold, model) jobs completed. The four Lichens 1 spatial-attention jobs first failed with an out-of-memory error and were rerun later with the same settings; the rerun results are used. Throughout, "our best model" is the attention variant with the highest mean balanced accuracy on the test units of that dataset, so the difference to the best baseline is optimistically biased in our favour.

A. Spectral track

Table II gives the spectral-track results. The excitation-token model (`ex_tokens_v2`) has the highest mean on Collagen Pepsin ($91.1 \pm 7.6\%$), the second highest on Lichens 2 (86.5% against 86.6% for the RBF SVM) and 98.8% on Lichens 1, where all strong models are at the ceiling (SpectralFormer, 99.5%). It does not beat the strongest tuned baseline on any dataset: the paired difference is $+0.8$ [-1.5, $+3.1$] points against logistic regression on Pepsin (Wilcoxon $p = 0.454$, 9 wins and 6 losses over 15 units), -0.1 [-1.0, $+0.8$] against the RBF SVM on Lichens 2, and -0.6 [-1.3, -0.1] against SpectralFormer on Lichens 1 (Holm-adjusted $p = 0.750$, with only four units). Averaged over the three datasets, `ex_tokens_v2` has the best mean rank (3.86), followed by the RBF SVM (3.97) and SpectralFormer (4.57); with three datasets the Nemenyi critical difference is 9.62 ranks, so no pair is distinguishable on this basis.

Table III gives the paired statistics against every baseline. On Pepsin, `ex_tokens_v2` is better than the RBF SVM, random forest, gradient boosting, k-NN, 1D-CNN and SpectralFormer with Holm-adjusted Wilcoxon $p \leq 0.031$, and than the MLP with $p_{\mathrm{W}}^{\mathrm{Holm}} = 0.051$. On Lichens 2 it is better than the random forest and the 1D-CNN after Holm correction ($p_{\mathrm{W}}^{\mathrm{Holm}} = 0.041$ and 0.016). The two tests disagree on how much this shows. Under the Nadeau–Bengio corrected t-test, which accounts for the overlap between training sets of different folds and repeats and is the conservative reference here, only the k-NN comparison on Pepsin ($p_{\mathrm{NB}} = 0.024$) and the 1D-CNN comparison on Lichens 2 ($p_{\mathrm{NB}} = 0.029$) fall below 0.05, and these p-values are not adjusted for the eight comparisons per dataset. The Wilcoxon test treats the 15 (or 10) units as independent although they are built from nine (or 28) distinct specimens, and is therefore anti-conservative; we do not treat the Wilcoxon results as evidence of generalisable superiority over the baselines it separates.

The other attention variants are below the best baselines (Table II). Measurement tokens, the reference prototype and the convolutional-stem hybrid are 5.2, 9.9 and 10.6 points below logistic regression on Pepsin, and 3.5, 3.2 and 4.2 points below the RBF SVM on Lichens 2; on Pepsin the differences to logistic regression of the prototype and the hybrid are significant by the Holm-adjusted Wilcoxon test (0.005 and 0.010) but not by the corrected t-test ($p_{\mathrm{NB}} > 0.17$). Within our family, tokenising whole emission spectra per excitation works, while tokenising single measurements does not, under this data regime. Because the family differs in several aspects at once (tokenisation, sequence length, embedding, search space), this is a comparison of members, not an isolating ablation.

B. Spatial track

Table IV reports the spatial track, with the same paired statistics in Table III. Spatial attention has the highest mean on Pepsin ($87.9 \pm 14.3\%$) but the difference to the patch-mean random forest, $+1.4$ [-6.1, $+7.7$] points over six units, is far inside the fold-to-fold spread (standard deviations of 10 to 18 points for all spatial models on Pepsin). It is below the strongest baseline on Lichens 1 (99.1% against 99.6% for the 2D-CNN) and Lichens 2 (85.6% against 87.1%, -1.5 [-3.6, -0.1] points, four losses in four units; Holm-adjusted Wilcoxon $p = 0.125$, the smallest value attainable with four units). The 2D-CNN is the strongest spatial baseline on Lichens 1 and Lichens 2. Spatial context does not improve over the best spectral-only model on Pepsin (best spatial mean 87.9% on six units against 91.1% for `ex_tokens_v2` on 15; the unit sets differ, so this is indicative only), and on Lichens 1 and Lichens 2 the best spatial and best spectral means differ by less than 1 point (99.6 vs 99.5 and 87.1 vs 86.6). We find no reliable benefit of spatial context or of spatial attention under whole-specimen evaluation.

C. Per-dataset comparison

Fig. 4 shows the distribution of the spectral-track results. Variation across units dominates variation across models: on

TABLE II
SPECTRAL TRACK: BALANCED ACCURACY (%), MEAN ± STD OVER (REPEAT, FOLD) UNITS. BEST PER COLUMN IN BOLD, SECOND BEST UNDERLINED. MODELS BELOW THE RULE ARE OUR ATTENTION FAMILY (EX TOKENS V2: EXCITATION TOKENS; ME-ATTN (MEAS.): MEASUREMENT TOKENS; SMARTATTN V0: REFERENCE PROTOTYPE). THE Δ ROW IS OUR BEST MODEL MINUS THE STRONGEST BASELINE, WITH A BOOTSTRAP 95% CONFIDENCE INTERVAL OF THE MEAN PAIRED DIFFERENCE.

Model	Collagen Pepsin	Lichens 1	Lichens 2
LR	90.3 ± 9.2	99.2 ± 0.3	83.3 ± 7.8
RBF-SVM	87.3 ± 8.1	99.3 ± 0.4	**86.6 ± 7.0**
RF	83.7 ± 9.0	98.6 ± 0.8	83.2 ± 7.4
HGB	84.7 ± 6.9	97.5 ± 3.5	84.6 ± 7.5
kNN	81.3 ± 8.9	91.8 ± 5.7	83.5 ± 7.0
MLP	83.2 ± 13.0	98.8 ± 0.6	85.6 ± 5.6
1D-CNN	79.2 ± 9.3	98.0 ± 1.1	81.6 ± 5.8
SpectralFormer	83.9 ± 10.3	**99.5 ± 0.1**	86.1 ± 6.3
SmartAttn v0	80.5 ± 12.6	98.8 ± 0.8	83.4 ± 6.9
ME-Attn (meas.)	85.2 ± 9.2	98.2 ± 1.3	83.1 ± 5.2
ex tokens v2	**91.1 ± 7.6**	98.8 ± 0.8	86.5 ± 5.8
hybrid conv attn	79.7 ± 11.2	95.5 ± 4.6	82.3 ± 7.1
Best ours − best baseline [95% CI]	+0.8 [−1.5, +3.1]	−0.6 [−1.3, −0.1]	−0.1 [−1.0, +0.8]
Strongest baseline	LR	SpectralFormer	RBF-SVM
Units n	15	4	10

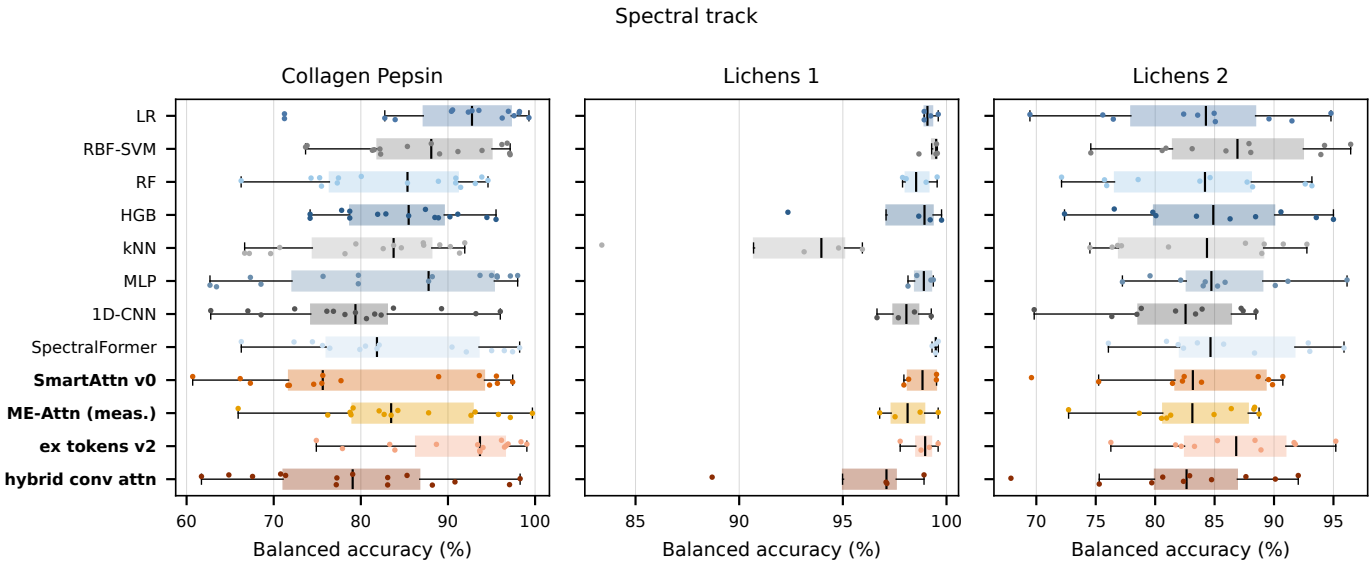


Fig. 4. Spectral track: distribution of balanced accuracy over (repeat, fold) units for every model, one panel per dataset (box: quartiles and median; points: units). Attention variants are in the lower four rows.

Pepsin the standard deviation over units is 7 to 13 points for all models, and the interquartile ranges of the top models overlap almost completely, so the ranking depends on which specimens are in the test folds (specimen heterogeneity). On Lichens 1 the standard deviations are below 1.5 points for all models except the k-NN, the hybrid and gradient boosting, and every strong model is above 98%; on Lichens 2 they are 5 to 8 points. Pixel-level metrics understate specimen-level performance: the specimen majority-vote accuracy of `ex_tokens_v2` is 95.6% on Pepsin (93.3% for logistic regression, the RBF SVM and gradient boosting) and 100% on Lichens 1 for every model, and 92.3% on Lichens 2 (best: k-NN, 96.5%), with large standard deviations given so few specimens. The class with only two regions in Lichens 2 contributes at most two test specimens per repeat, so its recall is estimated from very few independent regions.

D. Attention and feature importance versus wavelength selection

We compare attention-derived importance with the wavelengths selected by the unsupervised convolutional-autoencoder method of [13] (30 selected measurements for Pepsin and 13 for Lichens 1; no selection is available for Lichens 2). For each dataset and model we take the configuration of the sweep record with the highest inner-validation score, retrain it once on the outer-training specimens of repeat 0, fold 0 (with the same inner split for early stopping), and compute attention rollout and last-layer

TABLE III

PAIRED COMPARISON OF OUR ATTENTION VARIANTS AND BASELINES, ONE BLOCK PER DATASET AND TRACK. FIRST ROWS OF EACH BLOCK: EVERY ATTENTION VARIANT AGAINST THE STRONGEST BASELINE (HOLM ACROSS VARIANTS). REMAINING ROWS: OUR BEST VARIANT (MARKED *, SELECTED ON THE TEST UNITS, HENCE OPTIMISTIC) AGAINST EVERY OTHER BASELINE (HOLM ACROSS BASELINES). n: UNITS; $\bar{d}$: MEAN DIFFERENCE IN BALANCED ACCURACY IN POINTS WITH BOOTSTRAP 95% CI; W/T/L: WINS, TIES, LOSSES; p_{W}: WILCOXON p (UNADJUSTED), p_{NB}: NADEAU–BENGIO CORRECTED t-TEST p (UNADJUSTED), $p_{\mathrm{W}}^{\mathrm{Holm}}$: HOLM-ADJUSTED WILCOXON p. WITH $n = 4$ THE SMALLEST ATTAINABLE TWO-SIDED WILCOXON p IS 0.125.

Track	Dataset	Ours	vs.	n	$\bar{d}$ [95% CI]	W/T/L	p_{W}	p_{NB}	$p_{\mathrm{W}}^{\mathrm{Holm}}$
spectral	Collagen Pepsin	SmartAttn v0	LR	15	−9.9 [−14.3, −5.5]	2/0/13	0.001	0.172	0.005
spectral	Collagen Pepsin	ME-Attn (meas.)	LR	15	−5.2 [−10.5, +0.1]	4/0/11	0.095	0.536	0.189
spectral	Collagen Pepsin	ex tokens v2*	LR	15	+0.8 [−1.5, +3.1]	9/0/6	0.454	0.823	0.454
spectral	Collagen Pepsin	hybrid conv attn	LR	15	−10.6 [−16.2, −5.6]	4/0/11	0.003	0.221	0.010
spectral	Collagen Pepsin	ex tokens v2*	RBF-SVM	15	+3.8 [+1.2, +6.4]	12/0/3	0.010	0.359	0.031
spectral	Collagen Pepsin	ex tokens v2*	RF	15	+7.5 [+3.9, +11.0]	13/0/2	0.003	0.193	0.016
spectral	Collagen Pepsin	ex tokens v2*	HGB	15	+6.5 [+3.1, +9.6]	12/0/3	0.004	0.211	0.021
spectral	Collagen Pepsin	ex tokens v2*	kNN	15	+9.8 [+7.6, +12.6]	15/0/0	<0.001	0.024	<0.001
spectral	Collagen Pepsin	ex tokens v2*	MLP	15	+7.9 [+2.3, +14.1]	11/0/4	0.026	0.398	0.051
spectral	Collagen Pepsin	ex tokens v2*	1D-CNN	15	+11.9 [+7.5, +16.6]	14/0/1	<0.001	0.113	0.002
spectral	Collagen Pepsin	ex tokens v2*	SpectralFormer	15	+7.2 [+3.1, +11.6]	11/0/4	0.005	0.293	0.021
spectral	Lichens 1	SmartAttn v0	SpectralFormer	4	−0.7 [−1.3, −0.0]	1/0/3	0.250	0.329	0.750
spectral	Lichens 1	ME-Attn (meas.)	SpectralFormer	4	−1.3 [−2.2, −0.4]	1/0/3	0.250	0.237	0.750
spectral	Lichens 1	ex tokens v2*	SpectralFormer	4	−0.6 [−1.3, −0.1]	1/0/3	0.250	0.306	0.750
spectral	Lichens 1	hybrid conv attn	SpectralFormer	4	−4.0 [−8.6, −1.1]	0/0/4	0.125	0.326	0.500
spectral	Lichens 1	ex tokens v2*	LR	4	−0.3 [−1.0, +0.3]	1/0/3	0.375	0.619	1.000
spectral	Lichens 1	ex tokens v2*	RBF-SVM	4	−0.5 [−0.8, −0.1]	1/0/3	0.250	0.254	1.000
spectral	Lichens 1	ex tokens v2*	RF	4	+0.2 [−0.2, +0.7]	3/0/1	0.625	0.633	1.000
spectral	Lichens 1	ex tokens v2*	HGB	4	+1.3 [−0.7, +4.8]	1/0/3	0.875	0.646	1.000
spectral	Lichens 1	ex tokens v2*	kNN	4	+7.0 [+2.9, +12.7]	4/0/0	0.125	0.216	1.000
spectral	Lichens 1	ex tokens v2*	MLP	4	−0.0 [−0.5, +0.5]	2/0/2	1.000	0.992	1.000
spectral	Lichens 1	ex tokens v2*	1D-CNN	4	+0.8 [+0.5, +1.1]	4/0/0	0.125	0.064	1.000
spectral	Lichens 2	SmartAttn v0	RBF-SVM	10	−3.2 [−5.6, −1.0]	3/0/7	0.027	0.458	0.059
spectral	Lichens 2	ME-Attn (meas.)	RBF-SVM	10	−3.5 [−5.3, −1.7]	2/0/8	0.014	0.313	0.055
spectral	Lichens 2	ex tokens v2*	RBF-SVM	10	−0.1 [−1.0, +0.8]	6/0/4	1.000	0.950	1.000
spectral	Lichens 2	hybrid conv attn	RBF-SVM	10	−4.2 [−10.0, −0.7]	2/0/8	0.020	0.657	0.059
spectral	Lichens 2	ex tokens v2*	LR	10	+3.1 [+1.3, +4.9]	9/0/1	0.014	0.361	0.082
spectral	Lichens 2	ex tokens v2*	RF	10	+3.2 [+1.7, +4.6]	9/0/1	0.006	0.238	0.041
spectral	Lichens 2	ex tokens v2*	HGB	10	+1.9 [+0.6, +3.1]	8/0/2	0.020	0.432	0.098
spectral	Lichens 2	ex tokens v2*	kNN	10	+2.9 [+1.2, +4.7]	7/0/3	0.027	0.374	0.109
spectral	Lichens 2	ex tokens v2*	MLP	10	+0.9 [−0.1, +2.0]	6/0/4	0.275	0.656	0.826
spectral	Lichens 2	ex tokens v2*	1D-CNN	10	+4.9 [+3.8, +5.9]	10/0/0	0.002	0.029	0.016
spectral	Lichens 2	ex tokens v2*	SpectralFormer	10	+0.4 [−0.3, +1.2]	6/0/4	0.375	0.776	0.826
spatial	Collagen Pepsin	ME-Attn (spatial)*	Patch-mean RF	6	+1.4 [−6.1, +7.7]	4/0/2	0.688	0.862	0.688
spatial	Collagen Pepsin	ME-Attn (spatial)*	Patch-mean LR	6	+6.0 [−0.6, +13.1]	3/0/3	0.438	0.483	1.000
spatial	Collagen Pepsin	ME-Attn (spatial)*	2D-CNN	6	+4.9 [−3.7, +14.5]	3/0/3	0.688	0.656	1.000
spatial	Collagen Pepsin	ME-Attn (spatial)*	SpectralFormer-P	6	+3.5 [−6.7, +14.5]	5/0/1	0.312	0.767	1.000
spatial	Collagen Pepsin	ME-Attn (spatial)*	SSFTT	6	+2.8 [−7.4, +13.7]	2/0/4	1.000	0.821	1.000
spatial	Lichens 1	ME-Attn (spatial)*	2D-CNN	4	−0.6 [−1.0, −0.1]	0/0/4	0.125	0.254	0.125
spatial	Lichens 1	ME-Attn (spatial)*	Patch-mean LR	4	−0.3 [−0.8, +0.2]	2/0/2	0.625	0.581	1.000
spatial	Lichens 1	ME-Attn (spatial)*	Patch-mean RF	4	−0.3 [−0.9, +0.4]	2/0/2	0.625	0.695	1.000
spatial	Lichens 1	ME-Attn (spatial)*	SpectralFormer-P	4	+0.3 [−0.4, +1.0]	3/0/1	0.625	0.667	1.000
spatial	Lichens 1	ME-Attn (spatial)*	SSFTT	4	−0.5 [−1.4, −0.0]	1/0/3	0.250	0.504	1.000
spatial	Lichens 2	ME-Attn (spatial)*	2D-CNN	4	−1.5 [−3.6, −0.1]	0/0/4	0.125	0.577	0.125
spatial	Lichens 2	ME-Attn (spatial)*	Patch-mean LR	4	+1.6 [+0.3, +3.6]	4/0/0	0.125	0.548	0.625
spatial	Lichens 2	ME-Attn (spatial)*	Patch-mean RF	4	+1.8 [−0.2, +4.3]	3/0/1	0.250	0.593	0.625
spatial	Lichens 2	ME-Attn (spatial)*	SpectralFormer-P	4	+0.6 [−1.7, +2.5]	3/0/1	0.875	0.845	0.875
spatial	Lichens 2	ME-Attn (spatial)*	SSFTT	4	−0.5 [−0.8, −0.1]	0/0/4	0.125	0.396	0.625

attention averaged over up to 4000 pixels of the held-out test fold. Because this uses a single fold per dataset, the results below are descriptive and should be read as such; importance scores are not causal [31].

For the measurement-token model, the 30 measurements with the highest rollout importance on Pepsin include 13 of the 30 selected by the unsupervised method, against 5.7 expected by chance (hypergeometric $p = 0.0005$; Spearman correlation between importance and selection indicator $\rho = 0.48$). On Lichens 1 the overlap is 4 of 13 against 0.9 expected ($p = 0.007$, $\rho = 0.13$). On Pepsin the rollout concentrates on the 440 nm emission band under the 325, 340

TABLE IV
SPATIAL TRACK: BALANCED ACCURACY (%), MEAN ± STD OVER UNITS (6, 4 AND 4). PATCH SIZE P IS TUNED IN $\{5, 7\}$. BEST PER COLUMN IN BOLD, SECOND BEST UNDERLINED. THE Δ ROW IS SPATIAL ATTENTION MINUS THE STRONGEST BASELINE OF THE COLUMN, WITH A BOOTSTRAP 95% CONFIDENCE INTERVAL.

Model	Collagen Pepsin	Lichens 1	Lichens 2
Patch-mean LR	81.9 ± 15.4	99.4 ± 0.5	84.0 ± 3.9
Patch-mean RF	<u>86.4 ± 10.0</u>	99.3 ± 0.5	83.8 ± 4.5
2D-CNN	82.9 ± 13.1	**99.6 ± 0.2**	**87.1 ± 3.4**
SpectralFormer-P	84.3 ± 17.6	98.7 ± 1.3	84.9 ± 3.5
SSFTT	85.1 ± 14.8	<u>99.6 ± 0.6</u>	<u>86.0 ± 1.6</u>
ME-Attn (spatial)	**87.9 ± 14.3**	99.1 ± 0.8	85.6 ± 1.9
Best ours − best baseline [95% CI]	+1.4 [−6.1, +7.7]	−0.6 [−1.0, −0.1]	−1.5 [−3.6, −0.1]
Strongest baseline	Patch-mean RF	2D-CNN	2D-CNN
Units n	6	4	4

and 365 nm excitations and the selected measurements lie at the short-wavelength end of the emission range and in the long-wavelength tails (Fig. 5). The excitation-token model, by construction, assigns importance to whole excitations: its attention concentrates on the 415 nm excitation in both lichen datasets and on 325 nm (rollout) and 385 nm (last layer) in Pepsin. A band-level overlap with the selected measurements is not meaningful for it, since all emission bands of one excitation share one score, and we do not interpret the overlap values computed for it. The agreement of the measurement-token importance with the unsupervised selection is a coincidence of two label-free signals on two datasets, in a single fold; it does not show that either identifies the measurements that matter for classification, and we did not retrain classifiers on the top-ranked measurements.

VII. DISCUSSION

Under a protocol that removes specimen leakage and equalises tuning, the advantage of attention over strong, tuned classical baselines is small or absent on these data. The best attention variant, the excitation-token model, matches the best tuned baseline (paired differences -0.6 to $+0.8$ points, intervals that include zero on two of three datasets) and is better than the other neural models on Pepsin; it does not outperform logistic regression, the RBF SVM or SpectralFormer. The remaining variants are 3 to 11 points below the best baselines on the two datasets with room above the ceiling.

Why excitation tokens and not measurement tokens. We can state what the data show and offer hypotheses that we did not test. Every attention variant has the same training data: a few specimens per class, so the number of independent examples is very small compared with the number of pixels, and inter-specimen variance dominates (Fig. 4). Measurement tokens present 150–210 tokens that the model must aggregate, with a small budget (three configurations, at most 25 epochs); excitation tokens present 6–8 tokens, each a whole emission spectrum embedded by one linear layer, which makes the cross-excitation intensity relations that carry the class information directly available and constrains the model more. This is consistent with the development finding that per-excitation normalisation lowers accuracy by 4–8 points because the cross-excitation intensity ratios are class-informative (Sec. IV-F). The MLP, 1D-CNN and hybrid are also weaker than the linear model on Pepsin, which points to the amount of data and the tuning budget as limits for all flexible neural models, not to attention specifically. We did not run the wavelength-encoding ablations or the shuffled-wavelength control that would show whether the continuous wavelength encoding contributes anything beyond the intensity values; the results therefore do not support the claim that wavelength-aware encoding is the reason for the performance of the excitation-token model.

Statistical evidence. The units of analysis are correlated (Sec. VIII). Wilcoxon tests on units give small p-values against several baselines on Pepsin and Lichens 2, but the corrected test, which we regard as the reference, leaves essentially none significant; the fair reading is that the excitation-token model is at least as good as most baselines and probably better than k-NN and, on Lichens 2, than the 1D-CNN, and that no more can be concluded from 3 to 5 specimens per class.

Spatial context. Adding a pixel neighbourhood did not help under whole-specimen evaluation on any dataset (Table IV). We have not tested why; with a handful of specimens, spatial texture may be as specimen-specific as the spectrum, and the extra input adds capacity that the data cannot constrain.

Importance. Attention over measurement tokens agrees with an unsupervised band-selection method above chance on the two datasets where both exist, which indicates that both pick up the same dominant structure of the data (the excitation–emission regions with high variance), but this does not establish that those measurements are the most useful for classification.

Protocol versus model. On the development set, random-pixel splits gave 97.9–98.4% balanced accuracy and whole-specimen folds 86.8–88.3% for the same models, a gap of 10–12 points (Sec. V-A). This protocol effect is larger than the difference between our best attention model and the best baseline on every dataset (less than 1 point), and comparable to the spread between the best and the worst of the twelve spectral models on Pepsin (11.9 points); the protocol changes the measured accuracy more than the choice among the leading models does.

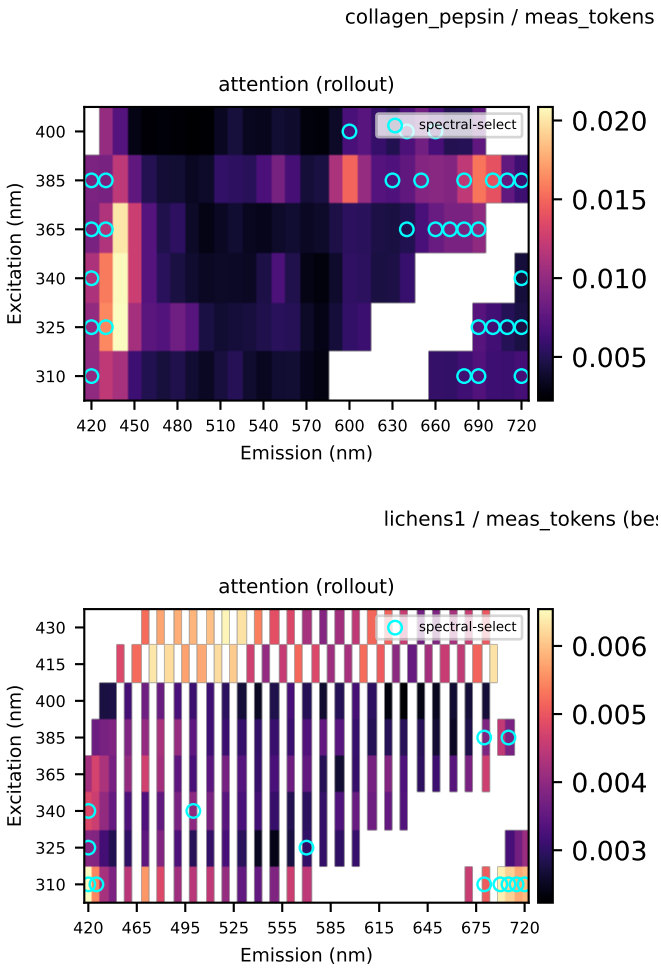


Fig. 5. Attention-rollout importance of the measurement-token model on the (excitation, emission) grid for one outer-training fold (repeat 0, fold 0; top: Collagen Pepsin, bottom: Lichens 1). Circles mark the measurements chosen by the unsupervised selection method [13]. White cells are absent measurements.

VIII. Limitations

The number of specimens is small: three or four per class for the collagen and Lichens 1 data, which limits the number of independent test units (15, 4 and 10 correlated (repeat, fold) units for the spectral track, derived from only 9, 16 and 28 distinct specimens), and corrected tests have low power at this size; with four units no Wilcoxon p-value below 0.125 is attainable. In Lichens 2, "specimens" are spatially disjoint regions of the same scene, and regions of one class may come from the same thallus or from the same imaging session; this is weaker than separate specimens, and one class has only two regions (about 400 pixels), so the two outer folds each hold one of them. All data come from one instrument (one source, one LCTF camera), and robustness to instrument change is not tested. Class labels come from region masks and may contain boundary errors. Search spaces are finite and the budget is small ($T = 3$ configurations, at most 25 epochs, 2–5 repeats); the planned budget ($T = 8$, 40 epochs, more repeats) was reduced because of compute constraints (a single Apple M4 Pro GPU shared with other jobs), and a larger budget could change the ranking, particularly for large neural models such as measurement tokens. The increase of the spectral-track repeats was decided after seeing the first two repeats, and Lichens 1 has a single repeat. Our best attention variant is selected on the test units, so reported differences in its favour are optimistic. Baselines other than SpectralFormer and SSFTT are our own re-implementations scaled to this data. No ablation of the wavelength encoding, Stokes-shift feature, relative bias or token dropout, no shuffled-wavelength control, and no retraining of classifiers on top-ranked measurements were run; the importance analysis uses a single outer-training fold per dataset. HybridSN, a 3D-CNN, linear SVM, extra trees, a band-token transformer and axial attention were implemented but not evaluated in the final sweep. Foundation models were not evaluated.

IX. Conclusion

We introduced a whole-specimen, class-covering, equal-budget evaluation protocol for supervised ME-HSI classification and used it to compare a family of wavelength-aware attention classifiers with 13 tuned baselines on three datasets and two tracks. Tokenising each excitation's emission spectrum gave the best model on Collagen Pepsin and a result within 0.1 points of the best baseline on Lichens 2, and was better than SpectralFormer on Pepsin; tokenising single measurements, the prototype and the hybrid were 3 to 11 points lower. No attention variant is reliably better than the strongest tuned linear, kernel or CNN baseline, and spatial context did not help. Attention importance of the measurement-token model agrees with an unsupervised band selection above chance. Because the protocol, not only the model, determines the measured accuracy (10–12 points on our development set), we recommend that ME-HSI classification results be reported with specimen-level splits and equal tuning budgets, and that claims of improvement over strong baselines be supported by corrected tests on specimen-level units.

Data and Code Availability

Code, search spaces, the evaluation harness and result files are available at [repository URL and commit]. The Lichens and Collagen data are on Zenodo [39], [40]; [confirm which datasets (Lichens 1, Lichens 2, Collagen Pepsin, Collagen Acetic) each record contains and whether Lichens 2 is released]. The unsupervised selection method used for comparison is described in [13].

References

[1] K. R. Murphy, C. A. Stedmon, D. Graeber, and R. Bro, "Fluorescence spectroscopy and multi-way techniques. parafac," *Analytical Methods*, vol. 5, no. 23, p. 6557, 2013.

[2] R. Bro, "Parafac. tutorial and applications," *Chemometrics and Intelligent Laboratory Systems*, vol. 38, no. 2, pp. 149–171, 1997.

[3] A. Croce and G. Bottiroli, "Autofluorescence spectroscopy and imaging: a tool for biomedical research and diagnosis," *European Journal of Histochemistry*, Dec. 2014.

[4] M. Monici, "Cell and tissue autofluorescence research and diagnostic applications," in *Biotechnology Annual Review*, 2005, pp. 227–256.

[5] Y. Gabrielyan, A. Hunanyan, S. Bezirganyan, L. Davtyan, A. Avetisyan, N. Sarvazyan, A. Butavyan, and V. Yeghiazaryan, "Comparative analysis of deep learning methods for classification of ablated regions in hyperspectral images of atrial tissue," *IEEE Access*, vol. 13, p. 35029–35047, 2025.

[6] A. Hunanyan, N. Ter-Petrosyan, F. Villarruel, T. Soghomonyan, N. Sarvazyan, A. Butavyan, and V. Yeghiazaryan, "Deep learning classification of ablated atrial tissue in multi-excitation hyperspectral images," in *2025 15th Workshop on Hyperspectral Imaging and Signal Processing: Evolution in Remote Sensing (WHISPERS)*, Nov. 2025, p. 1–5.

[7] N. Ter-Petrosyan, A. Hunanyan, F. Villarruel, N. Sarvazyan, A. Butavyan, and V. Yeghiazaryan, "Comparative analysis of multi-excitation hyperspectral image configurations for ablated atrial tissue classification," in *2025 15th Workshop on Hyperspectral Imaging and Signal Processing: Evolution in Remote Sensing (WHISPERS)*, Nov. 2025, p. 1–5.

[8] G. Lu and B. Fei, "Medical hyperspectral imaging: a review," *Journal of Biomedical Optics*, vol. 19, no. 01, p. 1–24, Jan. 2014.

[9] J. Liang, J. Zhou, Y. Qian, L. Wen, X. Bai, and Y. Gao, "On the sampling strategy for evaluation of spectral-spatial methods in hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 55, no. 2, p. 862–880, Feb. 2017.

[10] J. Nalepa, M. Myller, and M. Kawulok, "Validating hyperspectral image segmentation," *IEEE Geoscience and Remote Sensing Letters*, vol. 16, no. 8, p. 1264–1268, Aug. 2019.

[11] D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillera-Arroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schröder, W. Thuiller, D. I. Warton, B. A. Wintle, F. Hartig, and C. F. Dormann, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, p. 913–929, Mar. 2017.

[12] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, Sep. 2023.

[13] N. Meloyan and N. Sarvazyan, "Unsupervised deep learning for wavelength selection in multi-excitation fluorescence imaging," *Scientific Reports*, 2026, under minor revision.

[14] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 54, no. 10, p. 6232–6251, Oct. 2016.

[15] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-d deep learning approach for remote sensing image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 56, no. 8, p. 4420–4434, Aug. 2018.

[16] N. Audebert, B. Le Saux, and S. Lefevre, "Deep learning for classification of hyperspectral data: A comparative review," *IEEE Geoscience and Remote Sensing Magazine*, vol. 7, no. 2, p. 159–173, Jun. 2019.

[17] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "Hybridsn: Exploring 3-d–2-d cnn feature hierarchy for hyperspectral image classification," *IEEE Geoscience and Remote Sensing Letters*, vol. 17, no. 2, p. 277–281, Feb. 2020.

[18] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[19] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations (ICLR)*, 2021.

[20] D. Hong, Z. Han, J. Yao, L. Gao, B. Zhang, A. Plaza, and J. Chanussot, "Spectralformer: Rethinking hyperspectral image classification with transformers," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, p. 1–15, 2022.

[21] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, p. 1–14, 2022.

[22] D. Hong, B. Zhang, X. Li, Y. Li, C. Li, J. Yao, N. Yokoya, H. Li, P. Ghamisi, X. Jia, A. Plaza, P. Gamba, J. A. Benediktsson, and J. Chanussot, "Spectralgpt: Spectral remote sensing foundation model," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 8, p. 5227–5244, Aug. 2024.

[23] G. C. Cawley and N. L. C. Talbot, "On over-fitting in model selection and subsequent selection bias in performance evaluation," *Journal of Machine Learning Research*, vol. 11, no. 70, pp. 2079–2107, 2010.

[24] K. Zhou, Z. Liu, Y. Qiao, T. Xiang, and C. C. Loy, "Domain generalization: A survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, p. 1–20, 2022.

[25] J. Wang, C. Lan, C. Liu, Y. Ouyang, T. Qin, W. Lu, Y. Chen, W. Zeng, and P. Yu, "Generalizing to unseen domains: A survey on domain generalization," *IEEE Transactions on Knowledge and Data Engineering*, p. 1–1, 2022.

[26] I. Gulrajani and D. Lopez-Paz, "In search of lost domain generalization," in *International Conference on Learning Representations (ICLR)*, 2021.

[27] M. Tancik, P. P. Srinivasan, B. Mildenhall, S. Fridovich-Keil, N. Raghavan, U. Singhal, R. Ramamoorthi, J. T. Barron, and R. Ng, "Fourier features let networks learn high frequency functions in low dimensional domains," in *Advances in Neural Information Processing Systems*, vol. 33, 2020.

[28] P. Shaw, J. Uszkoreit, and A. Vaswani, "Self-attention with relative position representations," in *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, 2018, p. 464–468.

[29] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, "Exploring the limits of transfer learning with a unified text-to-text transformer," *Journal of Machine Learning Research*, vol. 21, no. 140, pp. 1–67, 2020.

[30] S. Abnar and W. Zuidema, "Quantifying attention flow in transformers," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, p. 4190–4197.

[31] S. Jain and B. C. Wallace, "Attention is not explanation," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 2019, p. 3543–3556.

[32] K. Jamieson and A. Talwalkar, "Non-stochastic best arm identification and hyperparameter optimization," in *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics (AISTATS)*, ser. PMLR, vol. 51, 2016, pp. 240–248.

[33] L. Li, K. Jamieson, G. DeSalvo, A. Rostamizadeh, and A. Talwalkar, "Hyperband: A novel bandit-based approach to hyperparameter optimization," *Journal of Machine Learning Research*, vol. 18, no. 185, pp. 1–52, 2018.

[34] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," *Journal of Machine Learning Research*, vol. 13, no. 10, pp. 281–305, 2012.

[35] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.

[36] C. Nadeau and Y. Bengio, "Inference for the generalization error," *Machine Learning*, vol. 52, no. 3, p. 239–281, Sep. 2003.

[37] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[38] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine Learning Research*, vol. 7, no. 1, pp. 1–30, 2006.

[39] N. Chilingaryan and N. Sarvazyan, "250707_4dhsi_lichens_palette," 2026.

[40] K. Ghahramanyan, "250929_collagen," 2026.